

Machine Learning Lab 2

Instance based learning - KNN

Throughout the lab sessions, we will continuously do experiments on the MNIST digits dataset. The ultimate goal is to build a handwritten digit recognition system using Python.

1. In this task, you will use instance based learning technique (KNN) to solve the problem of handwritten digit recognition. As for the dataset, we still use the subset of MNIST which we used in our previous labs. We will use `KNeighborsClassifier` from `sklearn` library to perform the task. You can see more details about how to use these methods from https://scikit-learn.org/stable/auto_examples/neighbors/plot_classification.html#sphx-glr-auto-examples-neighbors-plot-classification-py

You can play around with your classification result to see how accurate your results are. For example plot all the images that is labeled '2' by KNN and see how many of them are actually digit '2', similar to what you did in Lab1 Task3.